Water Quality Test Reports: Summary & Analysis

Vishnu Shivam Tower, Thakur Village, Kandivali East, Mumbai | Prepared 29-10-2026 | Reference standard: IS 10500:2012 (Drinking Water)

Bottom line: Both samples have **Total Coliform and E. coli present**, so **neither is fit for drinking** as tested. The borewell water is also very turbid (216 NTU against a limit of 1) and shows organic pollution (BOD 8 mg/L, COD 24.8 mg/L), which points to sewage or surface water getting in. The kitchen water is chemically acceptable apart from slightly high hardness and alkalinity. The bacterial contamination needs urgent disinfection and a retest.

1. Reports reviewed

Report A: Borewell Water	
Lab ID / ULR	VAL/2512/160070 / TC1193925000035715F
Laboratory	Varni Analytical LLP (NABL)
Collected	16-12-2025, by lab staff
Report date	22-12-2025
Parameters	21 (2 biological, 19 chemical)
Lab's verdict	**Does not pass IS 10500**

Report B: Kitchen Water (labelled "Water")	
Lab ID / ULR	VAL/2609/230075 / TC1193926000039075F
Laboratory	Varni Analytical LLP (NABL)
Collected	23-09-2026, sent by courier
Report date	26-09-2026
Parameters	17 (2 biological, 15 chemical)
Lab's verdict	**Not fit for human consumption**

2. Results side by side

Acceptable limit = IS 10500 desirable limit (as printed on the report). Permissible limit = the relaxation IS 10500 allows when no alternative source exists. Status: **OK** = within acceptable; **Above acceptable** = over acceptable but within permissible; **FAIL** = over permissible or not allowed at all.

Parameter	Unit	Acceptable	Permissible	Borewell (Dec-25)	Status	Kitchen (Sep-26)	Status
Biological							
Total Coliform	/100 ml	Absent	Absent	Present	**FAIL**	Present	**FAIL**
E. coli	/100 ml	Absent	Absent	Present	**FAIL**	Present	**FAIL**
Physical							
Odour	–	Agreeable	–	Disagreeable	**FAIL**	Agreeable	**OK**
Taste	–	Agreeable	–	Disagreeable	**FAIL**	Agreeable	**OK**
Colour	Hazen	5	15	10.00	**Above acceptable**	ND<1.0	**OK**
Turbidity	NTU	1	5	216.00	**FAIL (43× permissible)**	1.00	**At the limit**
pH	–	6.5–8.5	No relaxation	7.69	**OK**	7.49	**OK**
Conductivity	µS/cm	–	–	739.0	–	864.0	–
Chemical							
Total Dissolved Solids	mg/l	500	2000	714.00	**Above acceptable**	425.00	**OK**
Total Hardness (as $CaCO_3$)	mg/l	200	600	190.00	**OK**	210.00	**Above acceptable**
Total Alkalinity (as $CaCO_3$)	mg/l	200	600	265.63	**Above acceptable**	348.00	**Above acceptable**
Calcium	mg/l	75	200	30.06	**OK**	46.09	**OK**
Magnesium	mg/l	30	100	27.95	**OK (near limit)**	23.09	**OK**
Chloride	mg/l	250	1000	63.99	**OK**	62.64	**OK**
Sulphate	mg/l	200	400	15.74	**OK**	17.36	**OK**
Fluoride	mg/l	1.0	1.5	ND<0.05	**OK**	ND<0.05	**OK**
Silica	mg/l	–	–	0.97	–	2.24	–
Pollution indicators (borewell only; not covered by IS 10500)							
Total Suspended Solids	mg/l	–	–	178.00	**High**	Not tested	–
Chemical Oxygen Demand	mg/l	–	–	24.79	**Elevated**	Not tested	–
Biochemical Oxygen Demand	mg/l	–	–	8.00	**High**	Not tested	–
Oil & Grease	mg/l	–	–	ND<4.0	**OK**	Not tested	–
Dissolved Oxygen	mg/l	–	–	Not tested	–	Not tested	–

Score: Borewell has **6 failures** (Coliform, E. coli, odour, taste, turbidity, plus high BOD and TSS) and **3 results above acceptable** (colour, TDS, alkalinity). Kitchen has **2 failures** (Coliform, E. coli) and **2 above acceptable** (hardness, alkalinity), with turbidity right at the limit.

3. Analysis

3.1 Borewell water (December 2025)

- **Faecal contamination:** Coliform and E. coli are present, and BOD (8 mg/L) and COD (24.8 mg/L) are both raised. Clean groundwater normally has BOD below 2–3 mg/L. Together these point to **sewage or surface run-off reaching the borewell**, for example through a damaged casing, a missing sanitary seal, or a nearby drain or septic line.
- **Very high turbidity (216 NTU) and TSS (178 mg/L):** the water carries a lot of silt or suspended matter. That also explains the colour of 10 Hazen and the disagreeable odour and taste. Particles this fine shield bacteria from chlorine, so **the water must be filtered before it is disinfected**.
- **Minerals:** TDS of 714 mg/L is above the acceptable limit but well within the permissible 2000. Hardness (190) and calcium are within limits. Magnesium (27.95) is close to its limit of 30. Fluoride, chloride and sulphate are low, which is good.
- **Data check:** the TDS-to-conductivity ratio is 714/739 = 0.97. The usual range is 0.55–0.75, so the TDS figure is probably inflated by fine colloidal matter passing through the filter. It is worth querying with the lab or re-checking at the next test.
- **Suitability: not suitable for drinking or cooking.** After filtration and chlorination it may be used for flushing, gardening and washing vehicles.

3.2 Kitchen water (September 2026)

- **Bacterial contamination:** Coliform and E. coli are present. Every chemical result is low and close to municipal supply quality, so the contamination most likely enters **inside the building**. Possible routes are the underground or overhead storage tanks, a cross-connection with borewell lines, leaking suction pipes near drains, or a dirty kitchen tap or purifier outlet.
- **Hardness 210 mg/L and alkalinity 348 mg/L** are slightly above the acceptable limits but well within the permissible limits (600). This is not a health risk, but it causes scaling in geysers, kettles and RO units. Because alkalinity is higher than hardness, there is no non-carbonate (permanent) hardness and some sodium bicarbonate is present.
- **Turbidity 1.00 NTU** is exactly at the limit. It should be monitored because it often rises with tank sediment.
- **Data check:** calcium and magnesium give a calculated hardness of 115 + 95 = 210 mg/L as $CaCO_3$, which matches the reported value. The borewell report balances the same way (75 + 115 = 190). Both reports are internally consistent.
- **How the sample was sent:** this sample was **sent by courier** in a plastic bottle, and the report says "Sample drawn by: NA". Bacteriological samples should be taken in a **sterile bottle** after flaming or disinfecting the tap and letting it run, kept chilled, and tested within 24 hours. A positive result from a non-sterile or delayed sample can be a false alarm, but it must be treated as real until a properly collected sample shows otherwise.

3.3 Trend

The two reports are for different sources, so they cannot show a trend for one source. Bacteria were found in **both** sources nine months apart, though, which suggests an ongoing problem with storage or plumbing hygiene rather than a single event.

4. Additional test: Dissolved Oxygen (DO)

What it measures	Oxygen dissolved in the water. This is different from BOD and COD, which measure how much oxygen the pollutants in the water use up.
Method	IS 3025 (Part 38), Winkler azide modification, or a calibrated DO meter on site
Sampling	Fill a 300 ml glass BOD bottle completely with no air bubbles. Fix the oxygen on site with $MnSO_4$ and alkaline iodide-azide. Analyse within 8 hours. A courier sample is not acceptable.

Interpretation	IS 10500 sets no limit. Good water is typically at least 5–6 mg/L. Low DO together with high BOD (as in the borewell) confirms organic pollution. Low DO in stored water shows stagnation.
Recommendation	Include DO for both borewell and kitchen water in the next round of testing, with the lab collecting the samples.

5. Recommended actions

#	Action	Priority
1	Stop using borewell water for drinking or cooking. Use boiled water, or RO + UV water, for drinking until a clean retest.	**Immediate**
2	Clean and disinfect the underground and overhead tanks: drain, scrub, then super-chlorinate (about 5 mg/L free chlorine, 30 minutes or more of contact) and flush.	**Immediate**
3	Inspect the borewell casing, cap and sanitary seal, and nearby drains and septic lines. Make sure borewell lines cannot mix with drinking-water lines.	**Within 1 week**
4	Treat borewell water before use: sediment or pressure sand filter, then activated carbon, then chlorination. Keep residual chlorine at 0.2–0.5 mg/L at the taps.	**Within 1 month**
5	Retest about 7 days after disinfection. The lab should collect the samples in sterile bottles. Test Coliform and E. coli, turbidity, residual chlorine and DO, for both sources.	**After cleaning**
6	Optional: a softener or anti-scalant for hardness and alkalinity in kitchen water if scaling is a problem.	Low
7	Routine programme: bacteriological test monthly, full IS 10500 panel every 6 months, tank cleaning every 6 months.	Ongoing

Based on the test reports issued by Varni Analytical LLP (VAL/2512/160070 and VAL/2609/230075). Results apply only to the samples tested. This summary is a technical interpretation and does not replace the original reports.